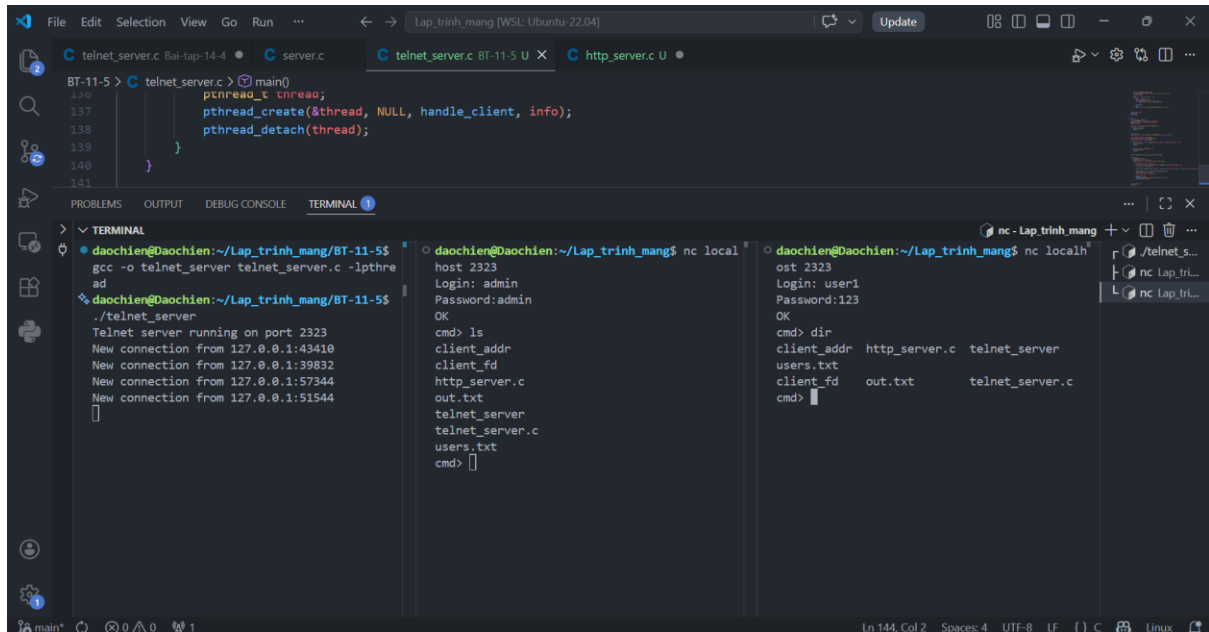


Đào Văn Chiến – 20225265

Bài tập Lập trình mạng 11/5

Bài 1: Dùng kỹ thuật đa luồng lập trình telnet_server



```
File Edit Selection View Go Run ... Lap_trinh_mang [WSL: Ubuntu-22.04] Update
telnet_server.c Bai-tap-14-4 server.c telnet_server.c BT-11-5 U http_server.c U
BT-11-5 > C telnet_server.c > main()
137 pthread_t tthread;
138 pthread_create(&tthread, NULL, handle_client, info);
139 pthread_detach(tthread);
140 }
141 }

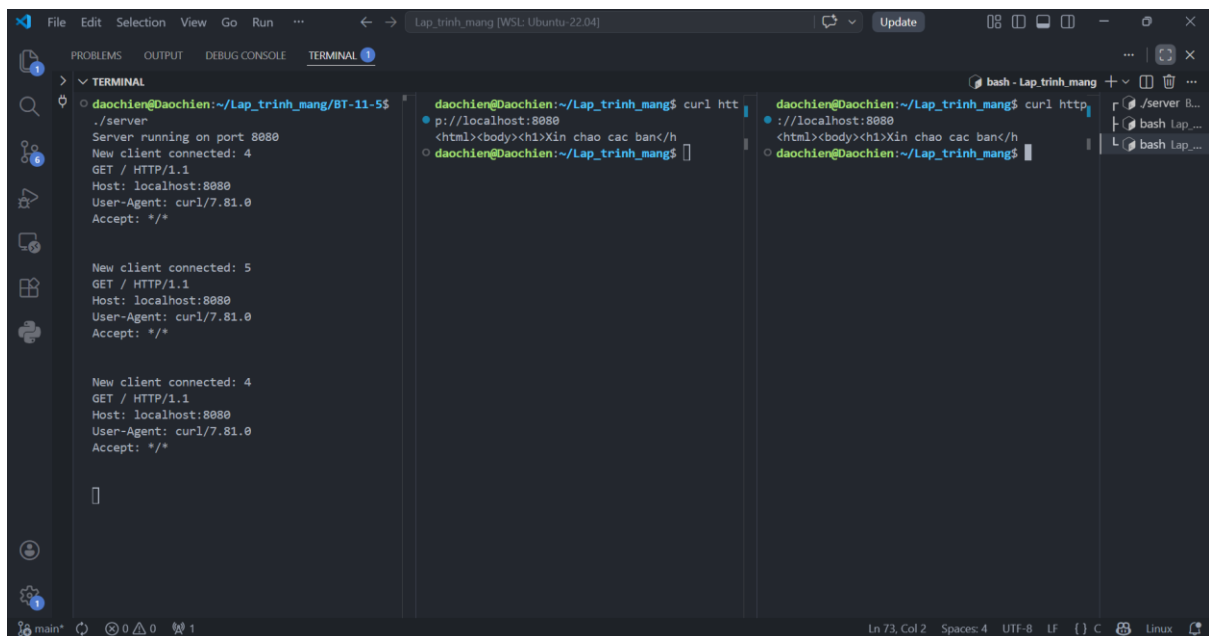
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1
daochien@daochien:~/Lap_trinh_mang/BT-11-5$ gcc -o telnet_server telnet_server.c -lpthread
daochien@daochien:~/Lap_trinh_mang/BT-11-5$ ./telnet_server
Telnet server running on port 2323
New connection from 127.0.0.1:43410
New connection from 127.0.0.1:39832
New connection from 127.0.0.1:57344
New connection from 127.0.0.1:51544

daochien@daochien:~/Lap_trinh_mang$ nc localhost 2323
host 2323
Login: admin
Password: admin
OK
cmd> ls
client_addr http_server.c telnet_server
client_fd out.txt telnet_server.c
cmd>

daochien@daochien:~/Lap_trinh_mang$ nc localhost 2323
ost 2323
Login: user1
Password:123
OK
cmd> dir
client_addr http_server.c telnet_server
users.txt out.txt telnet_server.c
cmd>
```

- Dùng đa luồng quản lý các client khác nhau

Bài2: Dùng kỹ thuật đa luồng lập trình http_server



```
File Edit Selection View Go Run ... Lap_trinh_mang [WSL: Ubuntu-22.04] Update
server.c http_server.c server.c
server.c
daochien@daochien:~/Lap_trinh_mang/BT-11-5$ ./server
Server running on port 8080
New client connected: 4
GET / HTTP/1.1
Host: localhost:8080
User-Agent: curl/7.81.0
Accept: */*

New client connected: 5
GET / HTTP/1.1
Host: localhost:8080
User-Agent: curl/7.81.0
Accept: */*

New client connected: 4
GET / HTTP/1.1
Host: localhost:8080
User-Agent: curl/7.81.0
Accept: */*

daochien@daochien:~/Lap_trinh_mang$ curl http://localhost:8080
<html><body><h1>Xin chao cac ban</h1></body></html>
daochien@daochien:~/Lap_trinh_mang$
```

Bai3: Lập trình ứng dụng time_server

```
File Edit Selection View Go Run ... Lap_trinh_mang [WSL: Ubuntu-22.04] Update
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
daochien@Daochien:~/Lap_trinh_mang$ cd BT-11
daochien@Daochien:~/Lap_trinh_mang/BT-11-S$ gcc -o time_server time_server.c -lpthread
daochien@Daochien:~/Lap_trinh_mang/BT-11-S$ ./time_server
Time server running on port 9090

daochien@Daochien:~/Lap_trinh_mang$ nc localhost
host 9090
GET_TIME dd/mm/yy
11/05/26

daochien@Daochien:~/Lap_trinh_mang$ nc localhost
ost 9090
GET_TIME mm/dd/yyyy
05/11/2026
Ln 114, Col 2 Spaces: 4 UTF-8 LF {} C Linux
```

- Sử dụng kỹ thuật đa luồng quản lý client và trả về định dạng thời gian theo format